



Pool boiling heat transfer characteristics of zircaloy and SiC claddings in deionized water at low pressure



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ABSTRACT

Since the Fukushima accident in 2011, deployment of silicon carbide (SiC) materials has been investigated as a replacement for zirconium-based cladding in light water reactor claddings. SiC cladding exhibits many advantages including an exceptionally high melting temperature, considerably slow rate of hydrogen generation from reaction with water at high temperatures, and less corrosion compared to currently used zircaloy cladding. To verify the enhanced safety margin and potential features of accident tolerance of the SiC cladding, the pool boiling heat transfer characteristics were investigated in deionized water under atmospheric pressure and compared to zircaloy-4 cladding. The SiC monolith claddings were manufactured and donated by GAMMA CTP. The zircaloy-4 claddings were heated via Joule heating using a 12 kW DC power supply while measuring the inner wall temperature. However, since the electrical resistance of SiC cladding is exceptionally high, indirect heating was applied by centering a heating element consisting of a stainless steel tube with an outer diameter of 6.35 mm inside the SiC cladding. The measured critical heat flux (CHF) of the SiC monolith cladding was approximately 63% higher than that of the zircaloy-4 cladding. Furthermore, after occurrence of the CHF, in spite of stable film boiling formation on the SiC surface, no physical damage was observed with the SiC cladding whereas the zircaloy-4 cladding experienced rapid physical degradation at the CHF and finally fragmented. High-speed video recorded at a frame rate of 1500 fps provided observation of the distinctive features of the characteristic boiling phenomena of the zircaloy-4 and SiC claddings. The hydrodynamic instability near the CHF, the transition from the CHF to film boiling, and the stable film boiling regime were captured successfully. For the first time, this study demonstrates that SiC cladding has the sustainable structural integrity with visual observation after the CHF occurrence and can be advantageous in securing a high safety margin for nuclear power reactor applications.

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1. Introduction

Zirconium alloys have been widely used as fuel cladding materials in commercial light water reactors (LWR) over the past several decades. They have various advantages as a nuclear fuel cladding due to their high mechanical strength, high melting point ($\sim 1850^\circ\text{C}$), high corrosion resistance, and low absorption cross-section for thermal neutrons [1]. However, despite these advantages, several limitations of zirconium-based claddings have been reported. Stress and corrosion induced cracking, embrittlement by irradiation, and hydrogen pickup may potentially damage fuel rods during operation due to extended burnup [2]. In addition, the exothermic oxidation process and hydrogen generation of a zircaloy cladding can be accelerated at high temperatures under

transition conditions, which makes the management of accidents more difficult [3].

In fact, approximately 1000 kg of hydrogen could have been produced from complete reaction of all of the claddings with high temperature steam in the Three Mile Island-Unit 2 (TMI-2) accident [4]. Although this did not occur in the TMI-2 accident, if the generated hydrogen was accumulated locally in the reactor system, it could detonate the reactor system, which was the case in Fukushima Daiichi nuclear power plant accident [5]. Recent numerical analysis using the MELCOR system code has also confirmed that the zircaloy and steam reaction is a significant heat source that needs to be managed for reactor vessel cooling in severe environments [6]. As such, demands for a higher safety margin and accident tolerant cladding have recently arisen.

As a strong candidate as an accident tolerant cladding material, silicon carbide (SiC) has received a lot of attention. SiC is the primary candidate for GFR, LFR, and VHTR, which are generation

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IV (Gen-IV) nuclear reactors [1,7]. SiC cladding exhibits noticeable advantages including its high thermal conductivity, high corrosion resistance, less waste disposal restrictions, and low neutron absorption cross-section [7]. Most of all, one remarkable advantage of adopting SiC cladding is its high melting point and the considerably lower rate of hydrogen generation in severe accident environments such that the enough amount of hydrogen explosion can be avoided before an accident management begins.

However, despite the reported advantages, important issues in using SiC cladding as a new nuclear fuel cladding remain unresolved. As practical issues, the manufacturability and cost competitiveness of SiC cladding have not been industrially achieved. Most importantly, there is a lack of phenomenological understanding of SiC cladding in terms of its reactor water chemistry and thermal–hydraulic operating conditions [1].

To address these important issues, several studies of SiC material have been conducted. Snead et al. evaluated the properties of SiC material under non-irradiated and irradiated conditions [8]. In their work, comprehensive data regarding SiC are presented, which is extremely valuable for nuclear applications. Carpenter investigated the irradiation performance of various SiC claddings in his Ph.D. dissertation work [9]. In the experiments, monolithic SiC and TRIPLEX tubes were tested under LWR conditions and the effects of irradiation and structural design on the SiC tubes were carefully investigated. From the experimental results, no significant creep and elastic deformation were observed over a large temperature range. However, SiC material experienced corrosion with water under LWR conditions. As a result, the strength of the SiC cladding was observed to be degraded. Carpenter discovered that these disadvantages could be improved using careful manufacturing techniques with high purity β -SiC and SiC fibers. He concluded that, in spite of its weaknesses, the SiC composite can be a feasible candidate for nuclear fuel cladding to overcome current issues of fuel claddings materials. More recently, Lee et al. experimentally assessed the SiC cladding oxidation rates under relevant conditions with loss-of-coolant accidents (LOCA) in order to understand its key failure mechanisms [3]. The experimental results showed noteworthy different oxidation behaviors of SiC and zircaloy-4. The SiC claddings underwent weight losses in steam while the zircaloy-4 experienced weight gains. They concluded that a reaction between silica (SiO_2) and water vapor (H_2O) caused the weight losses due to outward diffusion of the SiC product species. However, the oxidation rates of SiC were approximately three orders of magnitude slower than those of zircaloy-4. A more relevant study to the current investigation is the experimental work of aus der Wiesche et al., which involved the pool boiling heat transfer of diamond and SiC plate heaters [10]. The experimental results showed that thermal wall properties contributed to higher heat transfer coefficients of the diamond heater than those of the SiC heaters in the nucleate boiling regime. On the other hand, it was reported that the heating wall properties rarely affected the natural convection mode and bubble dynamics including the bubble departure diameter and frequency during nucleate boiling. Their study provides a useful reference on the pool boiling heat transfer of a SiC heater. However, understanding of the boiling crisis on a SiC surface remains insufficient because their experiments were limited to a maximum heat flux of 500 kW/m^2 due to the premature destruction of the oxidation layer at a high heat flux. Therefore, CHF data were unavailable in their study. Unlike the work of aus der Wiesche et al., a few studies reported CHF enhancement with SiC material using SiC nanofluids [11,12]. Besides the aforementioned experimental studies, a numerical study was also carried out for the potential application of SiC cladding in existing nuclear power plants. Seo et al. investigated the effects of the deployment of SiC claddings as a replacement of zircaloy-4 claddings on the thermal–hydraulic behavior of OPR1000

[13], which is a type of commercial power reactor in Korea. A large break loss-of-coolant-accident (LBLOCA) scenario was selected for the analysis using MARS code, which is a system safety analysis code developed in Korea. For simplification, the SiC cladding was modeled without changing the dimensions of the current zircaloy cladding. The simulation results showed lower peak cladding temperatures (PCT) for the SiC claddings than the zircaloy claddings in the blowdown and reflood phases. The reduced PCTs may be contributed to the higher thermal diffusivity of SiC, which enhances the safety margin during transient conditions.

Consequently, the objective of this study is to investigate the thermal–hydraulic behavior of SiC in pool boiling heat transfer including observation of CHF phenomenon. SiC monolith cladding heaters were selected and compared to zircaloy-4 cladding heaters. Pool boiling heat transfer experiments were performed in a pool of deionized water (DI water) at atmospheric pressure. The CHF and nucleate boiling heat transfer coefficients of both types of claddings were compared. Also, visualization of boiling was conducted using a high-speed video system to characterize the distinctive boiling regimes.

2. CHF correlation

Prediction of the CHF has been a crucial task in thermal–hydraulic evaluations to accurately estimate the safety margin for high power nuclear reactors. However, due to the complex thermal–hydraulic pressure, temperature, and flow characteristics, no single universal CHF prediction has been developed to date. Initiated with intuitive observations of fundamental pool boiling heat transfer, CHF prediction has a long history of development. In 1951, Kutateladze, a pioneer of the CHF field, developed a CHF correlation for the first time in the pool boiling of water on a plate heater, as shown in Eq. (1). Zuber (1959) later concretized the CHF triggering mechanism by introducing the instability theory, which states that a vapor jet velocity reaching the critical value triggers the CHF condition. Then, he determined the coefficient C in Eq. (1) to be $\pi/24$, which is the wavelength of the unstable vapor jet column repeatedly formed on the infinite plate heater. Since the pioneering work of Kutateladze and Zuber, Lienhard et al. also developed a similar CHF correlation on a plate heater with a slightly different C coefficient [14].

$$q''_{\text{crit}} = Ch_{fg}\rho_v^{1/2}[\sigma g(\rho_l - \rho_v)]^{1/4} \quad (1)$$

The development of CHF correlations has followed many directions depending on the heater geometry, orientation, and surface characteristics, to mention just a few factors [15,16]. In the current study, CHF with a tube geometry is of special interest and therefore, it is worth considering relevant CHF studies. In developing a CHF correlation for tube type heaters, many researchers extended the Zuber correlation as a reference frame. Sun and Lienhard (1970) developed the CHF correlation shown in Eq. (2) using the reference Zuber's prediction equation. The dimensionless radius shown in Eq. (3) was introduced to take into account the cylinder radius effect.

$$\frac{q''_{\text{crit}}}{q''_{\text{crit,Zuber}}} = 0.89 + 2.27 \exp(-3.44\sqrt{R'}) \quad (R' > 0.15) \quad (2)$$

The dimensionless radius R' is defined as follows:

$$R' = R\sqrt{\frac{g(\rho_l - \rho_v)}{\sigma}} \quad (3)$$

Tong and Tang (1997) stated that vapor removal configuration in the vicinity of the CHF depends on the heater diameter [17]. For small cylinders of $0.2 < R' < 2.4$, Sun and Lienhard's modified correlation is described by the empirical formula shown in Eq. (4).

$$q''_{\text{crit}} = 0.123 h_{fg} \rho_v^{1/2} \left[\frac{\sigma^3 g (\rho_l - \rho_v)}{R^2} \right]^{1/8} \quad (0.2 < R' < 2.4) \quad (4)$$

Lienhard and Dhir (1973) later confirmed this equation over the dimensionless radius range of $0.15 < R' < 1.2$. In addition, for large cylinders with $R' > 2.4$, Sun and Lienhard observed that the CHF becomes independent of the cylinder radius and developed the correlation shown in Eq. (5).

$$q''_{\text{crit}} = 0.118 h_{fg} \rho_v^{1/2} [\sigma g (\rho_l - \rho_v)]^{1/4} \quad (R' > 2.4) \quad (5)$$

A major Sun and Lienhard the CHF correlations by Kutateladze, Zuber, Sun and Lienhard, and Lienhard and Dhir was that the surface wettability has little impact on the CHF. However, some studies have revealed that surface wettability indeed has a great influence on the CHF. Improved surface wettability contributes to enhance the CHF and the contact angle of a liquid on the heater surface can be used as the indicator for the qualitative measurement of surface wettability. Liaw and Dhir (1986) considered the surface wettability effect on pool boiling CHF [18]. The resulting graphical solution is shown in Fig. 1 [16].

More recently, Kandlikar (2001) developed an empirical CHF correlation incorporating the surface wettability effect [19]. A recent study of pool boiling heat transfer by Kim et al. (2007) clearly shows that wettability has a great influence on the CHF [20]. Improved wettability contributes to enhanced CHF conditions and the degree of enhancement differs depending on the thermal–hydraulic pressure and flow conditions.

$$q''_{\text{crit}} = h_{fg} \rho_v^{1/2} \left(\frac{1 + \cos \beta}{16} \right) \left[\frac{2}{\pi} + \frac{\pi}{4} (1 + \cos \beta) \cos \phi \right]^{1/2} [\sigma g (\rho_l - \rho_v)]^{1/4} \quad (6)$$

In addition to the surface wettability effect, the roughness effect was investigated by Ramilison et al. (1992) as shown in Eq. (7) [21,22]. With a higher value of surface roughness, more micro-cavities are likely to be present and thus, enhanced nucleate boiling and thermal dissipation can be expected on a rougher surface [23,24].

$$q''_{\text{crit}} = 0.0366 (\pi - \beta_r)^{3.0} R_q^{0.125} \frac{\pi}{24} \left[h_{fg} \rho_v^{1/2} \{ \sigma g (\rho_l - \rho_v) \}^{1/4} \right]^{1/2} \quad (7)$$

Furthermore, many researches have revealed that enhanced boiling heat transfer and CHF could be obtained with porous media coated surfaces. Various mechanisms have been suggested to explain the enhancement with porous-layer surfaces in terms of

increased nucleation site density with micro-sized pores, reduced capillary pressure due to the porous wicking structure, or transport of liquid in interconnected flow paths [21,25,26]. Polezhaev and Kovalev (1990) investigated the porous media effect on the CHF, and suggested a correlation, considering the effects of porosity and pore size as shown in Eq. (8) [21].

$$q''_{\text{crit}} = 0.5 e^{2.28} h_{fg} \left[\frac{\sigma \rho_l \rho_v}{(\rho_l + \rho_v) R_{bg}} \right]^{1/2} \quad (8)$$

3. CHF experiments with zircaloy and SiC tubes

3.1. Description of experimental apparatus and heater design

3D drawings of the pool boiling facility used in this study are shown in Fig. 2. The water storage tank was assembled with polycarbonate walls and supported by stainless steel frames. The dimensions of the inner tank are $0.3 \times 0.3 \times 0.3 \text{ m}^3$. Since one of our major objectives in the pool boiling experiments was to visualize the boiling behavior for phenomenological understanding, three quartz windows were installed.

As mentioned earlier, two types of heaters were used in this study. As a reference case, a zircaloy-4 tube heater was prepared, which is the representative type of fuel claddings used in current commercial LWRs. Fig. 3 shows a schematic view of the zircaloy-4 tube heater. The zircaloy-4 tube was placed between two rectangular copper electrodes and mechanically connected by the cylindrical copper electrodes. Electrical power was directly supplied to the zircaloy tube through the electrodes. The inside of the tube was filled with epoxy for insulation and to prevent the inflow of water. In order to measure the wall temperature, three K-type TCs wrapped with metal foil were attached to the inner wall of the zircaloy tubes 90° azimuthally apart.

Fig. 4 shows a schematic view of the SiC tube heater. The design of the SiC heater assembly is rather complicated because direct heating of SiC was not viable and thus, an indirect heating method was employed. The heater assembly consisted of three major parts: an inner stainless steel grade 316 (SS316) tube with a 6.54 mm OD, an indium gap filler, and the monolithic SiC tube. The inner stainless steel tube plays the role of heat generation. The indium gap filler provides an opportunity for effective heat conduction generated by the inner SS316 tube. The heat generated from the inner heater element is conducted through the gap filler and outside surface of the SiC tube. As a result, the SiC heater assembly was designed very similarly to the design of an actual fuel rod. Both ends of the embedded SS316 tube were plugged mechanically with copper electrodes which were used to supply electrical power. Epoxy was solidified at both ends of the SiC tube in order to prevent inner indium from being leaked to the water pool due to its relatively low melting point of 157°C . In the experiments, temperature measurements for the SiC heaters were obtained at different positions. For the first heater sample, TCs were installed in the gap of the SiC heater while, for the second and third heaters, TCs were inserted in the center of the inner SS316 tube as shown in Fig. 4. The major dimensions of the zircaloy and SiC tubes are summarized in Table 1.

In addition to the description of heater design, it should be noted that different heating methods for the test heater were employed: direct heating for the zircaloy heater and indirect heating for the SiC heater. Due to the difference, there might be a limitation in comparison of boiling heat transfer and surface temperatures. However, the experiments are mainly focused on the CHF phenomenon and how the claddings change before and after the CHF. Furthermore, in general, to determine the maximum heat flux, applied power and heating surface area are the sufficient

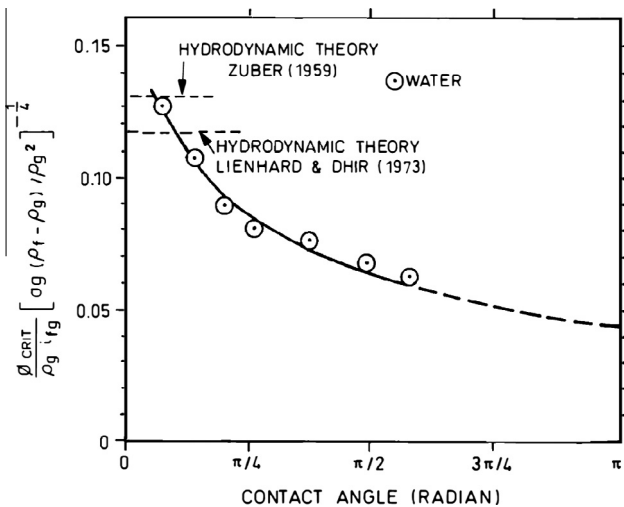


Fig. 1. CHF prediction as a function of the contact angle [16].

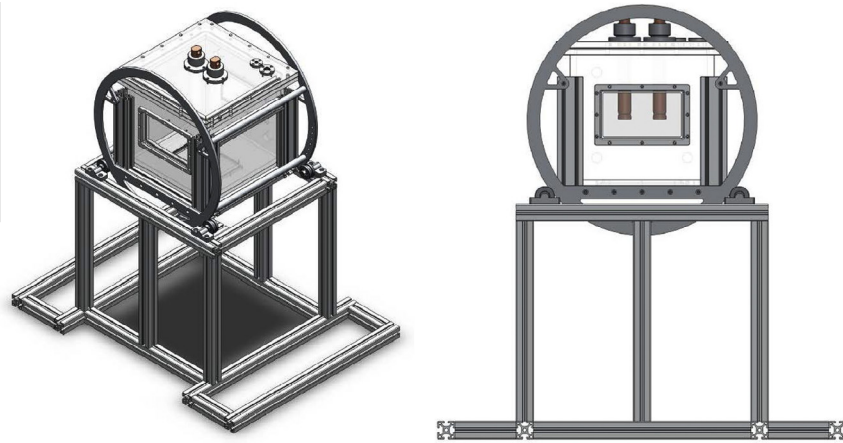


Fig. 2. 3D drawings of the pool boiling apparatus.

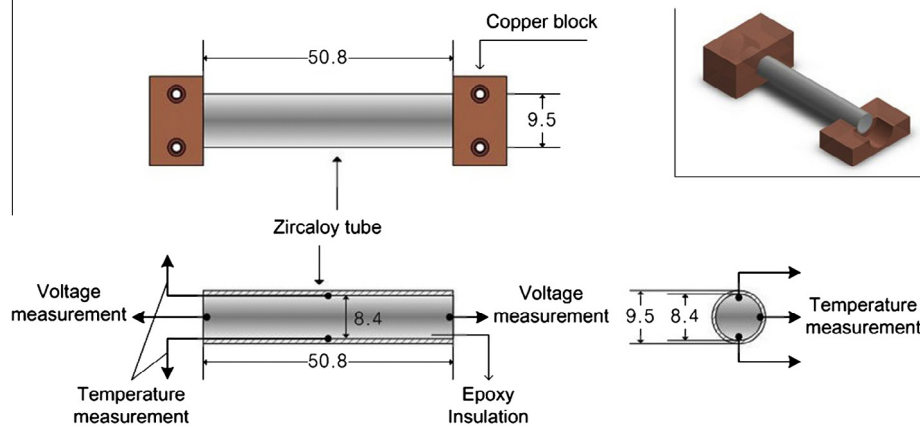


Fig. 3. Schematic of the zircaloy heater assembly.

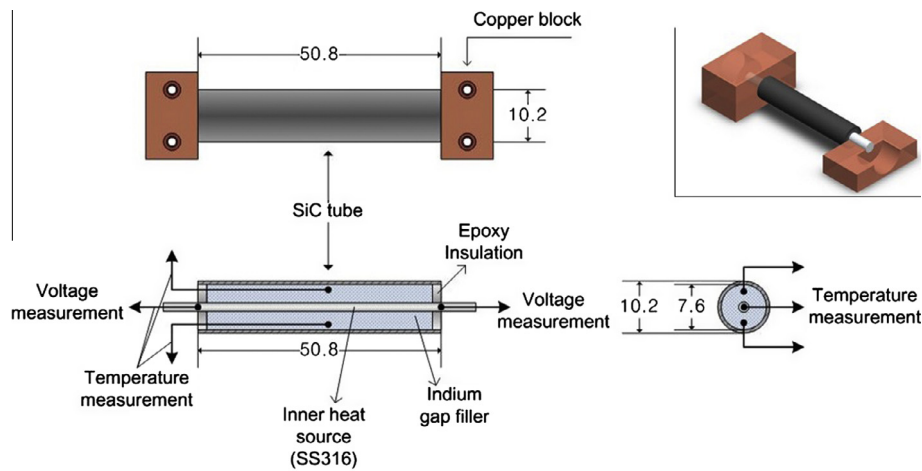


Fig. 4. Schematic of the SiC heater assembly.

Table 1
Dimensions of the zircaloy-4 and SiC tubes (mm).

	O.D.	I.D.	Thickness	Heated length
Zircaloy-4	9.5	8.4	0.6	50.8
SiC	10.2	7.6	1.3	50.8

parameters as many researchers carried out [26–28]. Moreover, unlike the design of zircaloy heaters, the TCs of SiC heaters were positioned in the gap with solidified indium for the first sample, and inserted in the inner SS316 tube with solidified epoxy for the second and third samples. Although it is sufficient to determine a CHF value with measured voltage and current signals, this eventually caused a hindrance to define accurate positions of TCs and

solve the conduction equation to obtain surface temperature as inner temperature increases highly. Considering the limitations in an experiment, results of CHF values and visual observations, especially before and after the CHF, is discussed in Section 4.

3.2. Test procedure

The pool boiling heat transfer experiment was conducted by applying the following procedures. A tube heater instrumented with a K-type TC and voltage taps was assembled to the copper electrodes and the corresponding heater assembly was placed in the pool of water. The pool of water was then heated using two immersion type heaters until boiling occurred at atmospheric pressure. While boiling the pool, the heater assembly was resistively heated using a 12 kW DC power supply (1200 A, 10 V). Starting with a very low current input, the electrical load was applied in a steady manner. At every heat flux increment of approximately 30 kW/m², a steady temperature and voltage were obtained. In addition, the boiling phenomena were captured using a high-speed video camera (Phantom V7.3) at a photographing rate of 1500 frames/s. Approaching the CHF, vigorous bubble departure and unstable vapor columns were observed. The CHF was determined to occur when a rapid temperature excursion was observed and/or by capturing the transition from nucleate boiling to film boiling. The CHF value was then determined as the average value between the previous step and current step which triggered the CHF. A typical power increase curve and detection of the CHF is shown in Fig. 5.

In a typical CHF experiment, the heater is often physically damaged and thus, transition to film boiling is rarely detected. In our study, however, the zircaloy and SiC heaters survived even upon triggering of the CHF and thus, visualization of the film boiling phenomena could be captured successfully using our HSV system. This visualization will be discussed in the experimental results section.

3.3. Uncertainty analysis

The uncertainty of the heat flux measurement was calculated using the propagation of error method [29]. Using the heat flux equation shown in Eq. (9), individual contributors which generate uncertainty can be determined. In this case, the contributors are the voltage, current, tube diameter, and tube length.

$$q'' = \frac{\text{Power}}{\text{Area}} = \frac{VI}{\pi DL} = V^1 I^1 \pi^{-1} D^{-1} L^{-1} \quad (9)$$

The uncertainties of the voltage, current, heater diameter, and length are less than 0.5%, 0.5%, 5%, 0.5%, and 0.2%, respectively. By applying the propagation of error method, Eq. (10) produces a final calculated measurement uncertainty of 6.6%.

$$\left(\frac{U_{q''}}{q''}\right)^2 = \left(\frac{U_V}{V}\right)^2 + \left(\frac{U_I}{I}\right)^2 + \left(\frac{U_D}{D}\right)^2 + \left(\frac{U_L}{L}\right)^2,$$

$$\frac{U_{q''}}{q''} = \left[(0.005)^2 + (0.005)^2 + \left(\frac{0.5 \times 10^{-3}}{9.5 \times 10^{-3}}\right)^2 + \left(\frac{2 \times 10^{-3}}{50.8 \times 10^{-3}}\right)^2 \right]^{1/2} = 0.0661 \quad (10)$$

3.4. Surface characterization

The contact angle is an indicator used to quantify the surface wettability effect, which is a dominant parameter affecting the boiling heat transfer characteristics [20]. As discussed in Section 2, wettability on a heating surface affects the CHF phenomenon and the contact angle was used as a major parameter of Kandlikar's CHF correlation [19]. Thus, a static contact angle measurement was performed to evaluate the wettability of the employed test heaters. Pictures of the contact angles on the zircaloy and SiC tubes are presented in Fig. 6. The apparent contact angles on the heaters were estimated to be 85° and 93° for the zircaloy and SiC tubes, respectively, which suggests that only a slight difference of 8° exists between the two heaters.

In addition to the contact angle measurements, the surface roughnesses of both heater surfaces were obtained. Surface roughness is also a key parameters which affects the nucleate boiling heat transfer since it can be closely related to the number of micro-cavities. There is a greater chance of increased boiling heat transfer for a surface with a higher roughness value [21,22]. As a result, the surface roughness may affect the CHF. The Ramilison's CHF correlation presented in Eq. (7) indicates that a higher CHF is expected for a rougher heating surface. Fig. 7 shows the measured surface roughnesses of the zircaloy and SiC heaters. The arithmetic averages of the absolute values of the surface feature heights for the zircaloy and SiC tubes, Ra, were 88 and 107 nm, respectively. The measured roughness of the SiC tube is about 22% higher than that of the zircaloy tube.

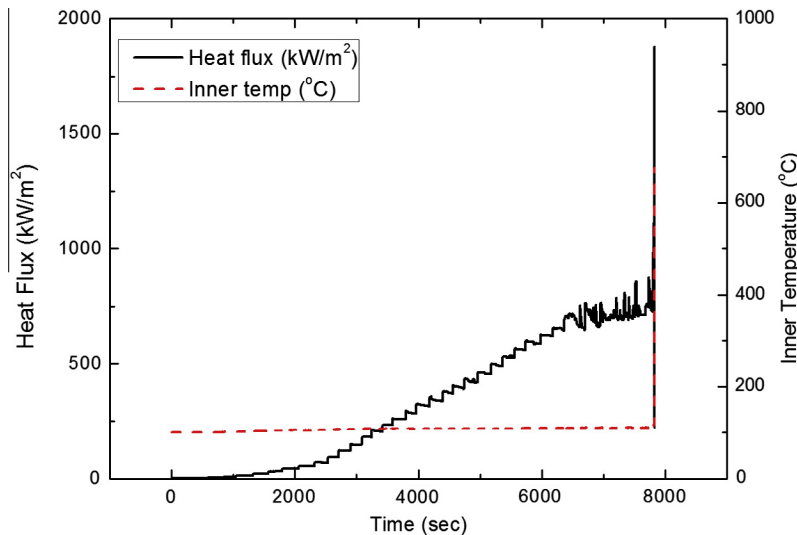


Fig. 5. The heat flux and inner wall temperature as a function of time for the SiC heater.

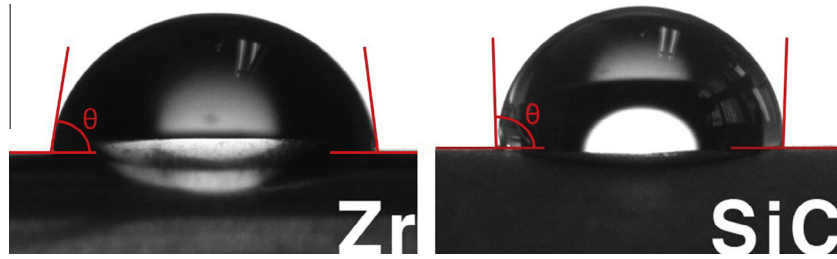


Fig. 6. Apparent contact angles: zircaloy: 85° (left) and SiC: 93° (right).

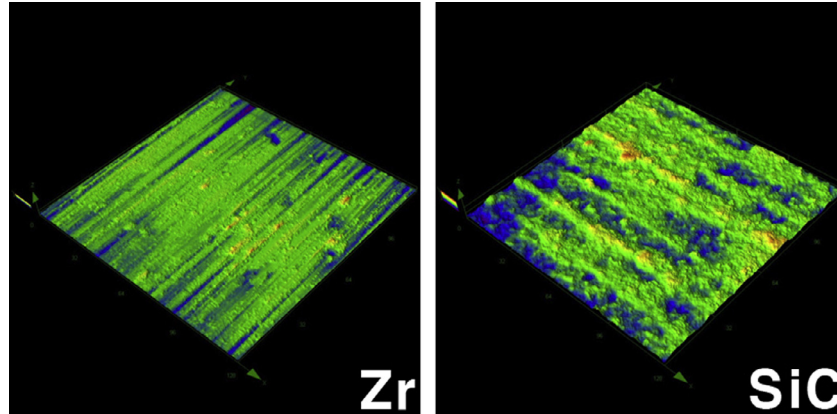


Fig. 7. Surface roughness, Ra (nm): zircaloy: 88 (left) and SiC: 107 (right).

4. Results and discussion

Boiling heat transfer experiments were carried out with increasing heat flux until the CHF occurred. Five zircaloy and three SiC heaters were prepared for comparison. In particular, for the two zircaloy samples, in addition to a heating test, a quenching test could also be conducted since the structural integrity of the heaters was maintained even after the CHF occurred. This is a rare case because for a typical CHF experiment, a metal heater experiences sudden rupture as soon as the CHF is triggered and thus, the experiments cannot continue. Fig. 8 shows the heat flux and inner wall temperature of both of the heaters as a function of time. The heat flux gradually increased in the range of 30–50 kW/m².

Normally, the tests were terminated after reaching the maximum temperature but in the same test, film boiling could be established by decreasing the heat flux as soon as the film boiling was established, as shown in Fig. 8.

4.1. Maximum and minimum heat fluxes of the zircaloy-4 cladding

In order to obtain reference values for this comparison study between the zircaloy and SiC claddings, CHF pool boiling heat transfer experiments of the zircaloy-4 cladding tube heaters were conducted in a pool of DI water under atmospheric pressure. The resulting CHF values obtained in the experiments are presented

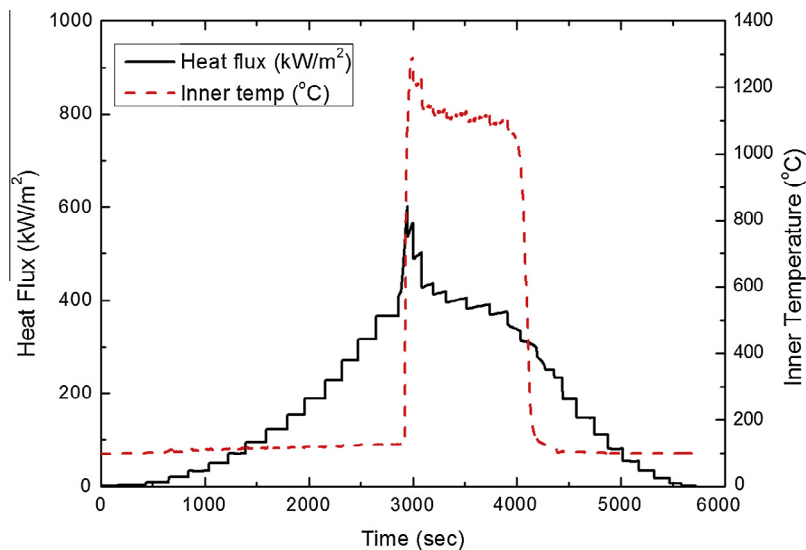


Fig. 8. The heat flux and inner wall temperature as a function of time for the zircaloy heater.

Table 2
Summary of the CHF values of the zircaloy-4 claddings (kW/m²).

Test no.	Test 1	Test 2	Test 3	Test 4	Test 5	Average
CHF	517	650	648	753	607	635

in Table 2. The average CHF or maximum heat flux value of the zircaloy-4 claddings was 665 kW/m².

It is of interest to compare the measured CHF values with existing predictions. The CHF could be explained by considering geometrical considerations and the surface wettability effect. Zuber's correlation (1959) shown in Eq. (1) is widely known as a conventional CHF correlation developed for an infinite flat heater. Here, the coefficient C is $\pi/24$ (≈ 0.13) and the Zuber's prediction results in a CHF value of 1,107 kW/m². However, since the test sample heaters have the geometry of a horizontal tube, comparison with other CHF models with geometrical considerations is more practical. Sun and Lienhard (1970) noticed that the diameter of the heater affected the CHF phenomenon [17]. In order to develop a CHF correlation most relevant to the tube geometry, they introduced a dimensionless heater radius, R' , as defined by Eq. (3). In the current study, the constant R' was evaluated to be 1.90 as the zircaloy heaters have a radius of 4.75 mm. The CHF correlation for small cylinders ($0.2 < R' < 2.4$) based on their analysis is suggested by Eq. (4). Furthermore, Lienhard and Dhir conducted extensive experiments to verify the equation using various liquids. Their model predicts a CHF value of 887 kW/m² for the zircaloy cladding. This result indicates that, for a small cylinder geometry, the CHF is predicted to be lower than the Zuber's hydrodynamic limit as the geometrical effect is considered. The difference between the measured CHF of the zircaloy cladding and predicted value is 222 kW/m². Although the measured CHF value is lower than the prediction, the tendency of decreasing CHF is observed in the Sun and Lienhard's correlation and the experimental results as compared to the Zuber's prediction.

Unlike the Zuber's correlation, which does not consider the heater tube geometry or surface effect, various researches have investigated the surface wettability effect. Liaw and Dhir (1986) observed that the CHF decreased with increasing contact angle in the range of 27–107°, as shown in Fig. 1. In their study, the CHF was evaluated by using the Zuber's model as a reference. The empirical results shown in Fig. 1 indicate values of 0.070, 0.068, and 0.064 for the constant C in the Zuber correlation when the contact angles are 70°, 80°, and 90°, respectively. It can be deduced that the CHF values predicted by Zuber and Lienhard–Dhir correspond to a low contact angle. Using the contact angle data (85°) of the current zircaloy heater, a CHF value of 556 kW/m² is predicted, which is almost half the value obtained using Zuber's

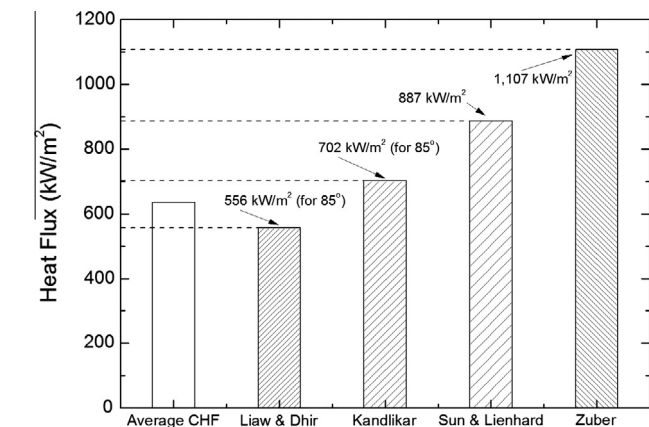


Fig. 9. CHF of the zircaloy heaters with suggested predictions.

prediction. This value is close to our measured CHF of 635 kW/m², as shown in Table 2.

More explicitly, Kandlikar's model (2001) incorporates the effects of the surface contact angle, as shown in Eq. (6). In this equation, β and ϕ are the surface contact angle and heater orientation angle, respectively. The Kandlikar's correlation predicts CHF values of 923, 775, and 630 kW/m² for contact angles of 70°, 80°, and 90°, respectively. Fig. 9 shows the experimental results of the zircaloy heaters with the suggested CHF predictions. The difference between the measured CHF and Kandlikar's model is only 67 kW/m². In this regard, it is believed that the measured CHF is reasonable and the experiments were conducted successfully.

In addition to the heat addition process used to determine the CHF, a reverse test of heat reduction for one zircaloy heater was carried out. This reverse process is believed to be difficult to utilize in CHF experiments since in general, after experiencing the CHF, a heater surface is damaged and breaks. Fortunately, in the quenching test of the sample, the test result indicates the minimum heat flux. Fig. 10 shows the boiling curve including both the points of maximum and minimum heat fluxes. This observation provides an opportunity to understand fundamental behavior in the boiling process. While the CHF point indicates that a stable film is developed on the heater surface, the minimum heat flux point shows where the vapor film collapses and nucleate boiling is reestablished [30].

4.2. Comparison of the CHF data of the zircaloy-4 and SiC claddings

Similar to the zircaloy-4 claddings, CHF and relevant boiling heat transfer experiments were conducted with the SiC monolith claddings. A summary of the CHF values of the SiC claddings is presented in Table 3. The resulting average CHF value is approximately 1,037 kW/m², which is 63% higher than the value of the zircaloy-4 claddings. In order to explain the higher CHF, two important aspects are considered, which are the thermo-physical properties and the surface characteristics of the SiC heaters. It is well known that a higher CHF can be obtained with an increase in the thermal conductivity [31]. It has been reported that a higher value of thermal conductivity induces enhanced lateral heat dissipation on a heating surface, and delays the formation of hot/dry spots during nucleate boiling [32]. Table 4 summarizes the thermo-physical properties of the zircaloy-4 and SiC claddings. The thermal conductivity (k) of the SiC cladding is about 29 times higher than that of zircaloy-4. Not only the higher thermal conductivity, but also the thermal diffusivity (α) of the SiC cladding, which is given by Eq. (11), is about 29 times greater than that of the

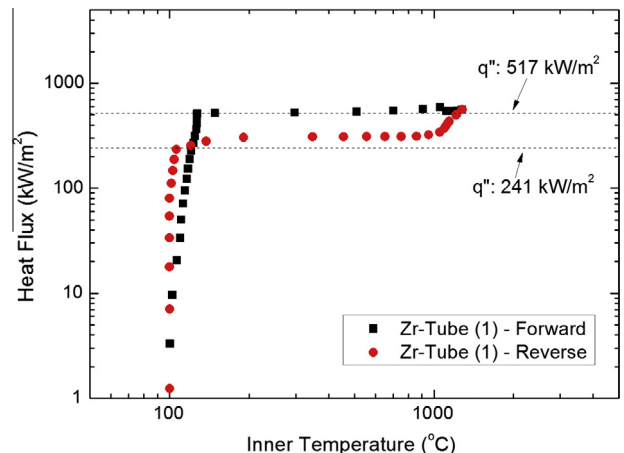


Fig. 10. Boiling curve of the zircaloy heater (no. 1 sample) with heating and the reverse process.

Table 3Summary of the CHF values of the SiC heaters (kW/m²).

Test no.	Test 1	Test 2	Test 3	Average	CHF _{SiC} /CHF _{Zr-4}
CHF	1105	899	1108	1037	1.63

Table 4

Thermo-physical properties of the zircaloy-4 and SiC claddings.

	ρ (kg/m ³)	k (W/m K)	C_p (J/kg K)	$\alpha \times 10^{-6}$ (J/m ² K)	T_{melt} (°C)
Zircaloy-4	6560	13.6	285	7.23	1850
SiC	3210	389.4	585	207.24	2797

zircaloy-4 cladding. Since the thermal diffusivity indicates how fast heat is transported in a material, this parameter can also attribute the thermal dissipation and the delayed formation of hot/dry spots.

$$\alpha = \frac{k}{\rho C_p} \quad (11)$$

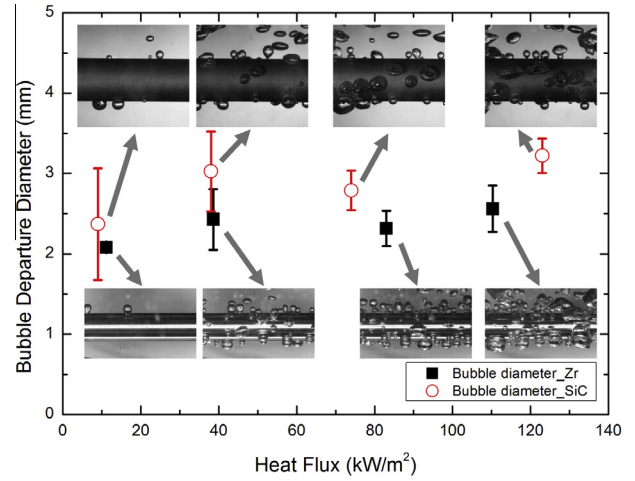
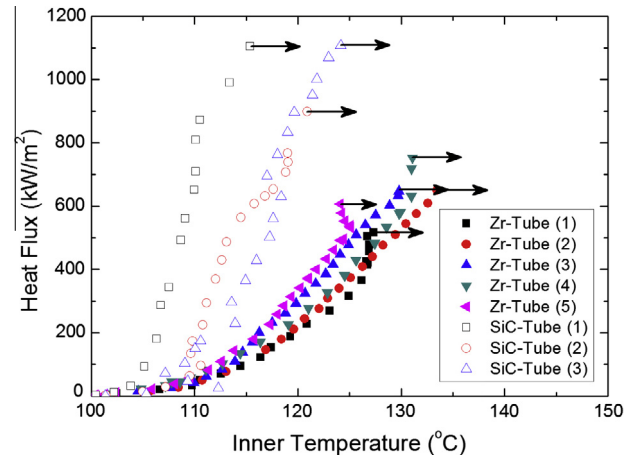
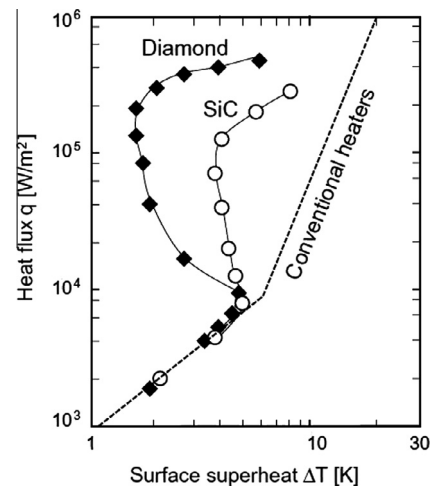
Ahn et al.'s recent study has reported a relevant observation and the effects of the thermal property on boiling performance and CHF. In their work, various thin graphene films with large thermal conductivity were used, and the CHF of the graphene-coated and bare heaters are compared. Furthermore, the local temperature profile was investigated using infrared (IR) high-speed visualization for observing the behavior of hot/dry spots. They concluded that improved CHF values were obtained for the graphene-coated samples, and the larger thermal conductivity of the graphene films inhibited the formation of hot spots [32].

In addition to the effects of the thermal property, roughness has a significant impact on the boiling heat transfer and CHF, which is known for the upper bound of the nucleate boiling regime [24,32]. Many researches have reported that the roughness is related to the number of nucleation sites and size of cavities, which could also affect bubble dynamics. The presence of larger unwetted cavities on the rougher surface resulted in lower superheats and enhanced thermal dissipation in nucleate boiling [23]. Moreover, as presented in Eq. (7), a higher value of CHF is obtained when the roughness increases since enhanced thermal dissipation could be occurred on a rougher surface. The surface characterization results in Section 3.4 shows 22% higher roughness of the SiC heaters and the relevant surface morphology. In this regard, the enhanced thermal dissipation is expected with a consideration of the boiling heat transfer and bubble behaviors. The following equation describes the performance of boiling heat transfer and relevant factors [32,33].

$$h \sim \frac{1}{A} \sum_{N_a} f^{1/2} \cdot D_d^2 \quad (12)$$

where N_a , f , and D_d are the nucleate site density, the bubble departure frequency, and the bubble departure diameter, respectively. Fig. 11 shows the measured bubble departure diameters with the applied heat flux, which is presented with the visualization results discussed in Section 4.3. The measured bubble departure diameters of the SiC heaters are about 24% higher than those of the zircaloy-4 heaters. Since no considerable difference in N_a and f is observed, it is evaluated that the boiling heat transfer of the SiC heaters is relatively higher, and thus these observation could imply the surface roughness effect as well.

Prior to further discussion, it should be noted that there were several limitations in the experimental results. To present the resulting boiling curves, the inner temperatures of the zircaloy and SiC heaters were used because surface temperatures of the heaters were not properly obtained. As mentioned in Section 3.1,

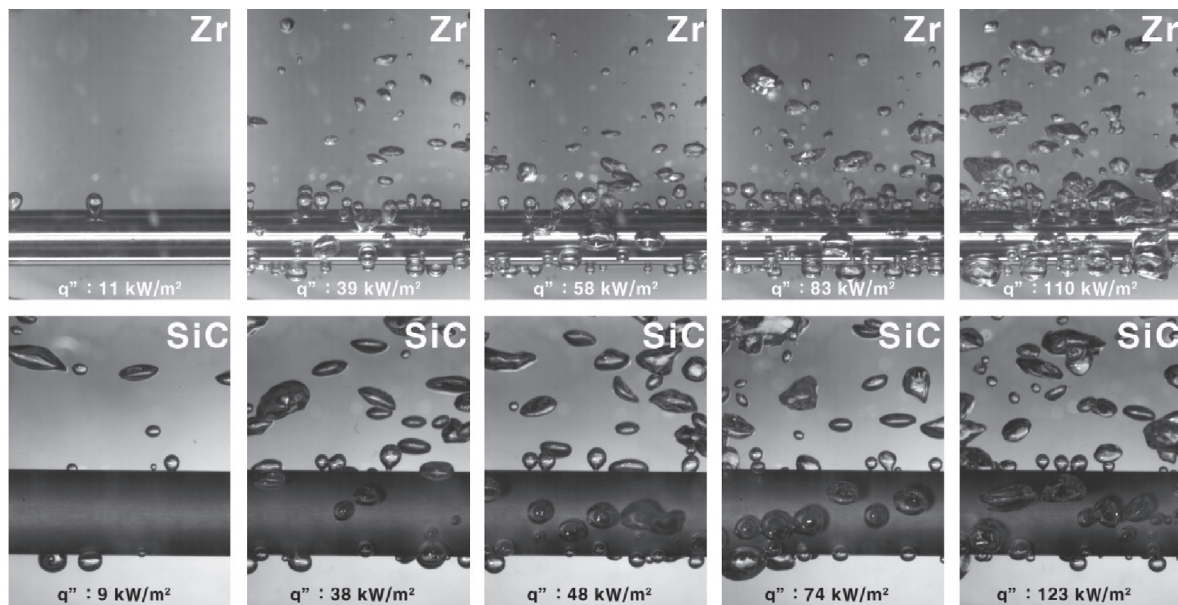
**Fig. 11.** Obtained values of bubble departure diameter with applied heat flux.**Fig. 12.** Pool boiling test results of the zircaloy and SiC heaters.**Fig. 13.** Boiling curves for diamond and SiC heaters [10].

unlike the case of zircaloy heaters, it is difficult to define precise positions of TCs in the SiC heater and solve the conduction equation. Due to the difficulties mentioned, the inner temperatures, which are measured at the positions of TCs installed, were selected to describe the boiling curves.

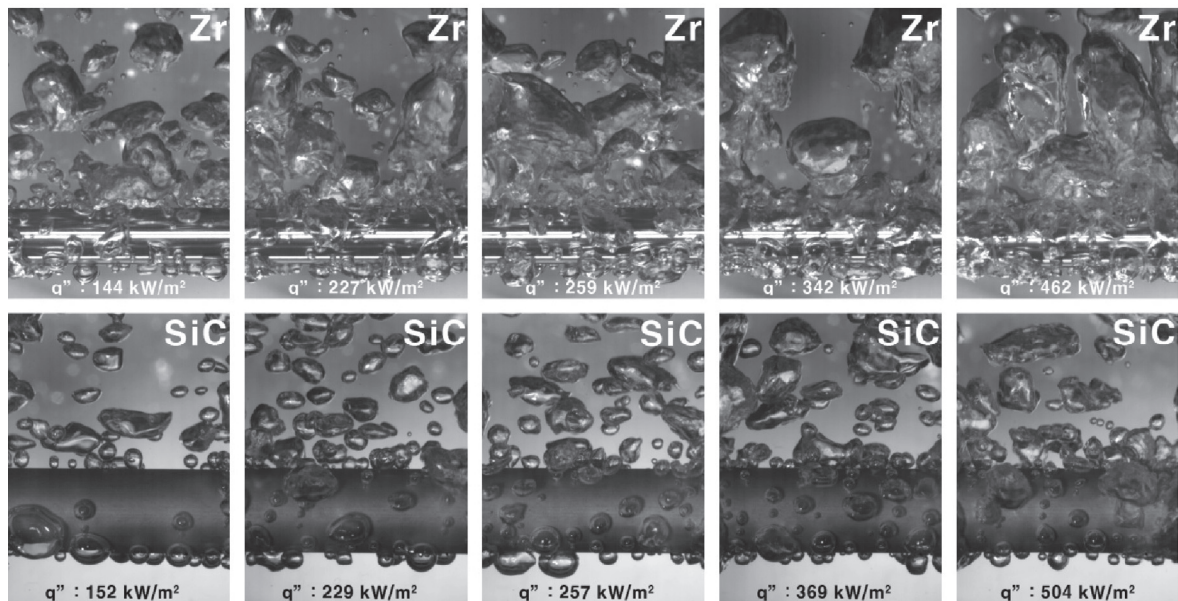
Fig. 12 shows the boiling curves where the applied heat flux is plotted as a function of the inner temperatures of the zircaloy and SiC heaters. As the higher CHF of the SiC heater indicates, the points at which departure from nucleate boiling occurs are in the upper region for the SiC heaters compared to the zircaloy heaters. The slopes of the SiC claddings are steeper than those of the zircaloy claddings. This implies that, given the same applied heat flux, the lower wall temperatures may be predicted for the SiC heaters. Although quantitative and accurate comparison is limited at the present stage, the experimental results with the bubble sizes of both heaters supports that higher boiling heat transfer occurs for the SiC heaters. Furthermore, a similar observation was reported by aus der Wiesche et al. [10]. They discussed that although the heater geometry is different with that of our study, a slower temperature variation of the heat flux was observed for the heater with a higher thermal conductivity, as shown in Fig. 13.

4.3. Visual observation of the zircaloy and SiC heaters

Visual observation of the pool boiling heat transfer characteristics of zircaloy and SiC heaters was conducted using a high-speed video system (HVS). The visualization work was focused on capturing the characteristic boiling phenomena of distinctive bubble dynamics during the nucleate boiling regime and the transition of boiling phenomenon from nucleate boiling to film boiling. Fig. 14 shows high speed images captured at a frame rate of 1500 fps for the zircaloy and SiC heaters during nucleate boiling until the CHF occurred. The boiling pictures show typical fundamental boiling phenomena summarized by Zuber. In the low and moderate heat flux regions, isolated bubbles were mainly observed for both heaters. As the heat flux was increased, nucleate boiling developed. In this region, as Zuber's hydrodynamic theory explains, each bubble combines with neighboring ones and a vapor column



(a) Low and moderate heat flux regimes



(b) high heat flux regime

Fig. 14. Boiling images as a function of heat flux for the zircaloy and SiC heaters.

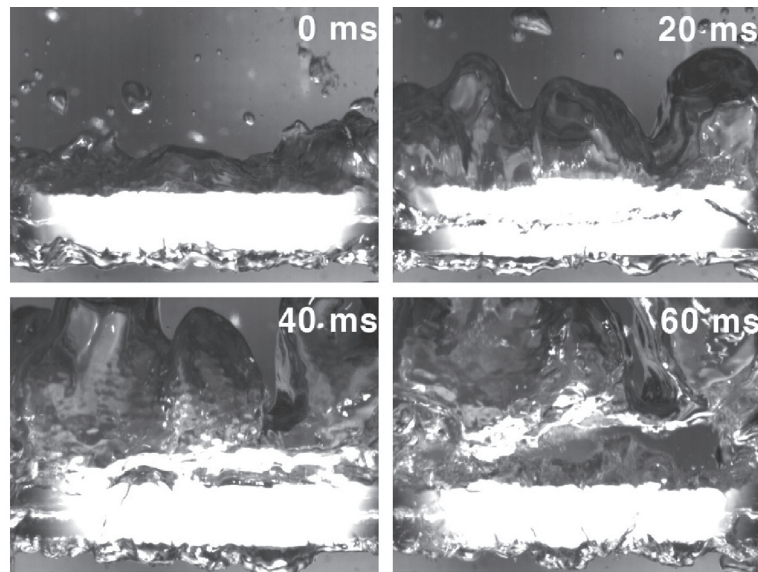
forms. However, as the heat flux was increased further up to $\sim 342 \text{ kW/m}^2$, a distinct difference in the boiling mode was observed. For the zircaloy heater, a previous vapor column merges with a new column and the heater surface is covered with vapor blankets. Finally, boiling crisis at the heater surface takes place in a very short time interval when most surfaces of the heaters are covered with the vapor blankets. On the other hand, for the SiC cladding, even at a high heat flux addition of about 500 kW/m^2 , isolated bubbles are still observed while the bubble size increases. Also, the bubble departure diameters from the SiC heater surface tend to be smaller than those from the zircaloy heater because isolated bubbles appear longer for the SiC heaters. As a result, the chance of vapor merge and coalescence for the SiC heater seems to be smaller than for the zircaloy heater and therefore, a higher CHF value is obtained.

It should be remarked that the melting temperature of the SiC cladding ($T_{\text{melt}} \sim 2800^\circ\text{C}$) is much higher than that of the heating element of the SS316 tube ($T_{\text{melt}} \sim 1400^\circ\text{C}$). Thus, before the surface temperature of the SiC cladding exceeds its melting temperature, the inner heater element was destroyed upon the CHF and the experiment was terminated. In a typical CHF experiment, the heater surface experiences a dramatic temperature excursion which destroys the heater. However, in the case of the SiC heater, despite the CHF occurrence, the structural integrity of every SiC

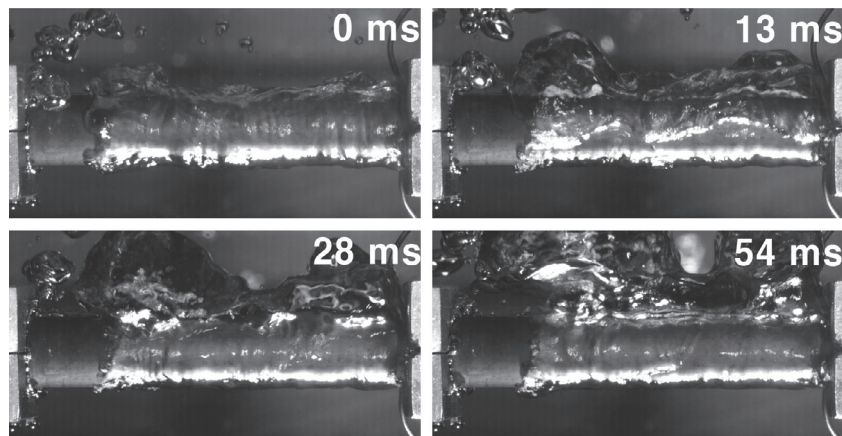
heater was maintained so that stable film boiling could be formed and observed following the CHF.

Fig. 15 shows a comparison of the boiling phenomena of the zircaloy cladding and SiC heaters at the instantaneous moment of CHF occurrence. For the zircaloy-4 cladding, near the CHF, unstable vapor columns were formed and the entire heater was covered by a crowd of vapors. Right after the CHF was triggered, the zircaloy-4 heater started to glow due to instantaneous thermal insulation by the vapor and physical rupture occurred. For the SiC cladding, similar hydrodynamic instability as the zircaloy heater was observed near the CHF condition. However, the SiC was thermally resistant to the high temperature and structurally intact even after the CHF where no physical damage occurred. As observed in Fig. 15(b), stable film boiling was maintained for several seconds because the SiC heater is thermally tolerant against the high temperature. However, the experiment was terminated because the inner stainless steel heater element melted. Moreover, due to the sudden break of the inner heater, heat was not supplied any more to the heating surface. Since the boiling images in Fig. 15(b) were recorded after the CHF occurred further, the left portion of the SiC heater was not covered by the vapor film.

One of the noticeable results in this experiment, in addition to the higher CHF with the SiC tube, is the structural integrity changes of both types of heaters after experiencing the CHF. In general, after



(a) CHF occurrence on the zircaloy heater



(b) CHF occurrence on the SiC heater

Fig. 15. Boiling behavior at the occurrence of CHF.

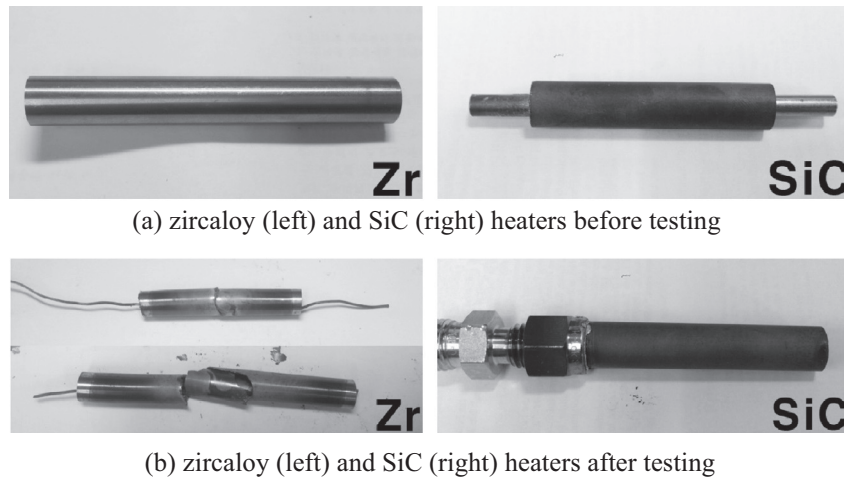


Fig. 16. Change of the structural integrities of the zircaloy and SiC heaters before and after experiencing the CHF.

CHF occurs, for most metallic materials such as stainless steel or zircaloy, the heating surface fails to maintain its structure. Since zircaloy rods are widely used in current nuclear reactors as a cladding material, preventing the CHF is an objective in different environments including normal operation as well as LOCA or severe accident conditions. Fig. 16 shows a visual comparison of the structural integrity for both types of heaters before and after the tests. The zircaloy claddings easily failed upon the CHF and were split into two or more fragments. On the other hand, the heater structure of the SiC claddings was maintained after the CHF although the film boiling was established around both surfaces, as shown in Fig. 15. As a candidate cladding material, this result implies that SiC has a great advantage in a nuclear power plant because adopting SiC material could provide relieved criteria in the reactor operation. Furthermore, from the aspect of the reactor safety in accident management, the feature of structural integrity with the CHF helps to allow more time to handle accidents since the cladding structure is a barrier for containing nuclear fuel fission fragments. Thus, the sustainable structure of SiC upon the CHF may provide an important advantage for nuclear reactor safety.

5. Summary and conclusions

In this study, pool boiling heat transfer experiments with zircaloy and SiC heaters were carried out. The CHF and nucleate boiling heat transfer of the zircaloy-4 and SiC cladding were compared. Specifically, sophisticated high-speed photographs of nucleate boiling, the CHF, and film boiling phenomena were captured. The major findings observed from this study can be summarized as follows:

- The experimental results showed a 63% higher CHF for the SiC heaters than the zircaloy heaters. This can be attributed to the high thermal conductivity and the higher surface roughness compared to the zircaloy-4 heaters.
- Although a quantitative comparison is limited somewhat, the boiling curves developed for the zircaloy-4 and SiC heaters suggest that higher nucleate boiling heat transfer is attainable with the SiC cladding, which could be due to the higher surface roughness and higher thermal conductivity compared to the zircaloy-4 cladding.
- Structural integrity of the SiC heaters was preserved even after experiencing the CHF whereas the zircaloy heaters failed and were fragmented. This demonstrates the superiority of the SiC cladding in terms of the improved safety margin for nuclear applications and excellent accident tolerance in post-LOCA con-

ditions. Thus, it is expected that, if accident tolerance of SiC claddings is confirmed in the future, adopting a SiC cladding may assure no hindrance in a steady-state operation in terms of CHF margin.

- Although the thermal superiority of the SiC cladding was demonstrated, more studies are needed to confirm the manufacturability and adaptability of the SiC suitable for nuclear applications. In particular, further experiments in more practical conditions are significant because material properties and surface characteristics could be effectively changed in real reactor environments. Moreover, resistance to mechanical friction and vibration of the SiC cladding have not been demonstrated, which should be properly addressed in future work.

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